

1. EDUCATION

- **School of Computer Science, Georgia Institute of Technology** Atlanta, GA, USA
PhD - Computer Science; GPA: 4.00 August 2025 - August 2030 (expected)
- **Faculty of Engineering, Alexandria University** Alexandria, Egypt
Bachelor of Science - Computer and Systems Engineering September 2016 - June 2021
Excellent with Degree of Honour (95.38%); ranked first

2. TECHNICAL SKILLS SUMMARY

- **Languages:** C, C++, Python, Java, SQL/PLSQL, Rust, Matlab, R, JavaScript, HTML, CSS
- **Frameworks:** PyTorch, TensorFlow, Keras, Spring Boot, Angular, Flask, Hibernate
- **Systems and Tools:** Linux, CloudLab, Git, GCP, Microsoft Azure, Firebase, ns-3, Android Studio, Oracle SQL Developer
- **Research Methods:** Distributed and networked systems, time-series analysis, deep learning, statistical evaluation

3. RESEARCH AND WORK EXPERIENCE

- **Georgia Institute of Technology (Georgia Tech)** Atlanta, GA, USA
Graduate Research Assistant (part-time) August 2025 - Present
 - **Current Research and Advisor:** Member of the Networks Research Group; advised by Prof. Ashutosh Dhekne since Summer 2026. Conducting research in scalable and efficient networked systems, sensing, and spatial intelligence.
 - **Prior Research and Advisor:** Previously advised by Prof. Ahmed Saeed from Fall 2025 through Spring 2026, focusing on datacenter systems, load balancing, overload control and low Earth orbit (LEO) satellite networks.
 - **Datacenter Load Imbalancing – USENIX NSDI 2026 Poster:** Proposed a load imbalancing strategy that concentrates requests on a subset of active servers while preserving idle machines as immediately available burst capacity. The design exploits sublinear server load–power scaling to save energy without reducing peak cluster capacity, unlike DVFS, or incurring the provisioning delay of server deallocation and autoscaling. The system proposes a distributed fast-reacting overload control method to ensure tail-latency SLOs are met. Our evaluation for *memcached* service on a 20-node CloudLab cluster shows up to **3.7–6.4%** reduction in CPU power while preserving throughput and satisfying the tail-latency SLO.
Implementation: Built a C-based prototype and compared multiple imbalanced allocations against load-balanced baselines using the *performance* and *schedutil* CPU governors.
 - **Cross-Source LEO Network Usage Measurement – ACM LEO-NET 2026 Paper (to appear):** Co-developed a framework for detecting high-confidence shifts in Starlink usage without private operator telemetry. Combined APNIC visible-user estimates, M-Lab speed-test activity, and Cloudflare Radar HTTP-traffic trends; retaining candidate events only when independent sources exhibited temporally aligned changes in the same direction. Case studies in Jamaica, Ukraine, and Germany identified shifts associated with emergency response, deployment changes, and service restrictions while filtering a prominent APNIC-only pattern that lacked support from the other vantage points.
Implementation: Implemented a Python time-series pipeline using EWMA, piecewise-linear segmentation, normalized slope-angle merging, and cross-source matching.
- **The American University in Cairo (AUC)** Cairo, Egypt
Researcher (full-time); joint research with Carnegie Mellon University March 2022 - August 2025
 - **Supervision and Research Scope:** Worked with Prof. Moustafa Youssef (AUC) and Prof. Khaled A. Harras (CMU) on indoor localization, mobile computing, context-aware systems, wireless sensing, and applied machine learning for smart environments.
 - **Lightweight Cellular Floor Identification – ACM SIGSPATIAL 2023:** Designed *UniCellular*, a smartphone-accessible system that identifies a user’s floor using only serving-cell signal strength available through standard Android and iOS APIs. Developed a compact CNN that improved accuracy by **3×** while reducing model size by more than **10×**. Built the Android data-collection application, Firebase multi-device storage pipeline, and TensorFlow/Keras training and inference workflow. *The paper received the ACM SIGSPATIAL 2023 Best Paper Runner-Up award.*
 - **Data-Efficient Floor Identification – ACM TSAS 2024:** Extended the floor-identification system with GANs and diffusion models that synthesize training data from a small number of real cellular samples. Reduced real-world data-collection effort by approximately **70%** while maintaining accuracy; the work was invited to the ACM TSAS Special Issue on the Best Papers of SIGSPATIAL 2023.
 - **Language-Model Transfer – IEEE/ACM SEC 2024:** Adapted a BERT encoder through domain-knowledge transfer to process cellular signal sequences rather than natural language. Implemented the PyTorch training and inference pipeline and achieved near-perfect floor-identification accuracy while retaining compatibility with consumer smartphones.
 - **Multimodal Mobility Sensing – IEEE VTC2023-Spring & ACM SIGSPATIAL 2024:** Designed CNN-LSTM and MLP pipelines that combine automatically learned cellular representations with domain-specific hand-engineered features for transportation-mode recognition. Evaluated multiple feature-fusion strategies and improved accuracy by **23%**. Developed Android applications for collecting cellular, inertial, and GPS signals and implemented the TensorFlow/Keras training pipeline.
 - **Domain Adaptation – IEEE TVT 2024:** Introduced denoising-autoencoder domain adaptation to address distribution shifts caused by phone placement and hardware heterogeneity. Enabled models trained on one placement and device type to generalize across unseen settings, improving robustness by **60%** over prior methods.

- **LEO Positioning Vision – ACM SIGSPATIAL 2024:** Proposed using private LEO satellite networks to complement GPS and enable ubiquitous global positioning, including in environments where conventional satellite positioning is weak, such as indoors. Identified the networking, sensing, systems, and deployability challenges required to realize the vision.
- **Human-as-a-Sensor – ACM SIGSPATIAL 2025 & IEEE Pervasive Computing 2026:** Proposed the Human-as-a-Sensor paradigm, in which humans act as intelligent multimodal sensing agents that complement traditional sensors. Leveraged advances in brain-computer interfaces to motivate extracting contextualized insights from brain signals for human-intelligence-augmented sensing and proactive spatial awareness.
- **Indoor Localization Survey – IEEE Access 2025:** Authored a survey of indoor localization systems in multi-floor environments, consolidating prior work and the open challenges in accurately identifying both horizontal position and floor level.
- **Research Mentorship:** Mentored six Alexandria University students in collaboration with Osaka University and two AUC students in collaboration with the University of Waterloo on context-awareness and ML for 5G-network simulation. Guided research methodology, literature review, experimentation, and academic writing, contributing to one accepted IEEE WINCOM 2025 paper and two accepted ACM SIGSPATIAL 2025 papers.

- **The American University in Cairo (AUC)** Cairo, Egypt
Teaching Assistant - Advanced Machine Learning and Computer Networks February 2022 - December 2024
 - **Responsibilities:** Led office hours and discussion sessions; prepared and graded assignments on networking and modern ML topics, including large language models; and advised students on practical course projects.
- **EJADA Systems Ltd.** Alexandria, Egypt
Junior Software Developer (full-time) August 2021 - March 2022
 - **Software Development:** Contributed to enterprise customer-relationship-management systems: used PL/SQL for Oracle database logic, Spring Boot to build RESTful APIs, and Angular to develop interactive forms and dynamic reports.
 - **Performance Optimization:** Investigated database-transaction bottlenecks and reduced execution time by more than an order of magnitude through bulk processing and database indexing.

4. RESEARCH PUBLICATIONS

Earlier publications appear under the author name Sherif Mostafa.

- **Sherif Abdelrazek**, Dhyey Thummar, and Ahmed Saeed. 2026. When Server-Level Load Imbalance Helps: Power Savings without Capacity Loss. In the Poster Session of the 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI '26). USENIX Association, 2 pages. *Posters page*.
- Stepan Kravtsov, **Sherif Abdelrazek**, Vaibhav Bhosale, and Ahmed Saeed. 2026. Keeping Up with the Trends: Identifying Shifts in LEO Satellite Usage Patterns. In the 4th Workshop on LEO Networking and Communication (LEO-NET '26). ACM, 7 pages. *To appear*. DOI.
- **Sherif Mostafa**, Khaled A. Harras, and Moustafa Youssef. 2026. Human-as-a-Sensor: Toward Autonomous Multimodal Sensing for Self-Determined Technology Engagement. *IEEE Pervasive Computing*, 25(1), 19–28. DOI.
- **Sherif Mostafa**, Khaled A. Harras, and Moustafa Youssef. 2025. Human-as-a-Sensor: Harnessing Brain Signals for Intelligent Multimodal Sensing in Spatially-Aware Systems. *ACM SIGSPATIAL '25*, 5 pages. DOI.
- **Sherif Mostafa**, Khaled A. Harras, and Moustafa Youssef. 2025. A Survey of Indoor Localization Systems in Multi-Floor Environments. *IEEE Access*, 13, 146396–146432. DOI.
- **Sherif Mostafa**, Moustafa Youssef, and Khaled A. Harras. 2024. Ubiquitous and Low-Overhead Floor Identification with Limited Cellular Information. *ACM TSAS. Invited to the Best Papers Special Issue*. DOI.
- **Sherif Mostafa**, Moustafa Youssef, and Khaled A. Harras. 2024. Accurate and Ubiquitous Floor Identification at the Edge using a Single Cell Tower. *IEEE/ACM SEC '24*, 206–219. DOI.
- **Sherif Mostafa**, Moustafa Youssef, and Khaled A. Harras. 2024. ModeSense: Ubiquitous and Accurate Transportation Mode Detection using Serving Cell Tower Information. *ACM SIGSPATIAL '24*, 184–195. DOI.
- **Sherif Mostafa** et al. 2024. Vision: Leveraging Low Earth Orbit Satellites for Future Ubiquitous Positioning. *ACM SIGSPATIAL '24*, 473–476. DOI.
- **Sherif Mostafa**, Khaled A. Harras, and Moustafa Youssef. 2024. Robust and Ubiquitous Mobility Mode Estimation Using Limited Cellular Information. *IEEE Transactions on Vehicular Technology*, 74(1), 1310–1321. DOI.
- **Sherif Mostafa**, Khaled A. Harras, and Moustafa Youssef. 2023. UniCellular: An Accurate and Ubiquitous Floor Identification System using Single Cell Tower Information. *ACM SIGSPATIAL '23*, Article 85, 1–10. *Best Paper Runner-Up*. DOI.
- **Sherif Mostafa**, Khaled A. Harras, and Moustafa Youssef. 2023. Ubiquitous Transportation Mode Estimation using Limited Cell Tower Information. *IEEE VTC2023-Spring*, 1–5. DOI.

5. SELECTED AWARDS

- **2024:** IEEE ComSoc competition honorary mention for ranking among the top 16 projects worldwide; ACM TSAS invitation to the Special Issue on the Best Papers of ACM SIGSPATIAL 2023; complimentary one-year ACM professional membership.
- **2023:** ACM SIGSPATIAL Best Paper Runner-Up award.
- **2021:** Professor Naim Aboutaleb Award for ranking first in the Faculty of Engineering, Alexandria University.

6. PROFESSIONAL ACTIVITIES

- **Reviewer:** IEEE INFOCOM 2023–2024, IEEE PerCom 2025, IEEE MDM 2024–2025, IEEE TMC, ACM TSAS, and ACM MobiCom 2023.
- **Member:** IEEE Graduate Student Member and IEEE Communications Society Graduate Student Member (2024–2027); ACM Professional Member (2024–2025); ACM SIGSPATIAL Member (2023–2024).
- **Volunteer:** Student volunteer at ACM SIGSPATIAL 2023.